Session 3: Orbit Transfers and Delta-V

Task 1: Explain why both Hohmann burns are purely tangential, and why a plane change is cheapest at apoapsis.

Tangential Hohmann Burns

- **Geometric Alignment**: The Hohmann transfer ellipse is strictly tangent to both the initial circular orbit (at periapsis) and the final target circular orbit (at apoapsis)
- **Directional Efficiency**: Because the orbits touch tangentially at these two impulse points, the orbital velocity vector of thr circular orbit and the velocity vector of the transfer ellipse point in the exact same physical direction.
- **Magnitude-Only Adjustment**: Consequently, both impulse maneuvers (Δv_1 and Δv_2) are purely tangential. They only modify the magnitude of the velocity vector without wasting propellent on redirecting the velocity direction:

$$\Delta v_1 = v_{trans,peri} - v_{circ,1}$$

$$\Delta v_2 = v_{circ,2} - v_{trans,apo}$$

Plane changes at apoapsis:

- The Δv cost for a pure inclination change Δi is given by $\Delta v_{plane} = 2v \sin\left(\frac{\Delta i}{2}\right)$
- Because $v = \sqrt{\mu\left(\frac{2}{r} - \frac{1}{a}\right)}$ is at its absolute minimum at apoapsis $\left(r_{apogee}\right)$, performing plane alignment at apoapsis drastically reduces propellant mass requirements relative to periapsis maneuvers.

Task 2: Hohmann Transfer

For Earth orbital calculations, $\mu = 398600\ km^3/s^2$, and $R_E = 6378\ km$.

1. **LEO** $(500\ km)$ **to GEO transfer**

- **Initial & Final Radii:** $r_1 = 6378 + 500 = 6878\ km, r_2 = 6378 + 35786 = 42164\ km$
- **Orbital velocities:** $v_1 = \sqrt{\frac{\mu}{r_1}} = \sqrt{\frac{398600}{6878}} = 7.613\,\frac{km}{s}, v_2 = \sqrt{\frac{\mu}{r_2}} = \sqrt{\frac{398600}{42164}} = 3.075\,\frac{km}{s}$
- **Transfer Semi-major Axis:** $a_{trans} = \frac{r_1 + r_2}{\mathbf{2}} = \frac{6878+42164}{2} = 24521.5\ km$
- **Transfer Velocities:**
  - $v_{t1} = \sqrt{\mu\left(\frac{2}{r_1} - \frac{1}{a_{trans}}\right)} = \sqrt{398600 \cdot \left(\frac{2}{6878} - \frac{1}{24521.5}\right)} = 9.983\ km/s$
  - $v_{t2} = \sqrt{\mu\left(\frac{2}{r_2} - \frac{1}{a_{trans}}\right)} = \sqrt{398600 \cdot \left(\frac{2}{42164} - \frac{1}{24521.5}\right)} = 1.628\ km/s$
- **Impulses & Time:**
  - $\boldsymbol{\Delta v_1} = v_{t1} - v_1 = 9.983 - 7.613 = 2.37\ km/s$
  - $\mathbf{\Delta v_2} = v_2 - v_{t2} = 3.075 - 1.628 = 1.447\ km/s$
  - $\boldsymbol{\Delta v_{total}} = \Delta v_1 + \Delta v_2 = 2.37 + 1.447 = 3.817\ km/s$
  - **Transfer duration:** $t_{trans} = \pi\sqrt{\frac{a_{trans}^3}{\mu}} = \pi\sqrt{\frac{24521.5^3}{398600}} = 19107.409\ s \approx 5.3\ h$

2. LEO $(400\ km)$ **to MEO** $(20200\ km)$

- **Initial & Final Radii:** $r_1 = 6378 + 400 = 6778\ km, r_2 = 6378 + 20200 = 26578\ km$
- **Orbital velocities:** $v_1 = \sqrt{\frac{\mu}{r_1}} = \sqrt{\frac{398600}{6778}} = 7.669\,\frac{km}{s}, v_2 = \sqrt{\frac{\mu}{r_2}} = \sqrt{\frac{398600}{26578}} = 3.873\,\frac{km}{s}$
- **Transfer Semi-major Axis:** $a_{trans} = \frac{r_1 + r_2}{2} = \frac{6778+26578}{2} = 16678\ km$
- **Transfer Velocities:**
  - $v_{t1} = \sqrt{\mu\left(\frac{2}{r_1} - \frac{1}{a_{trans}}\right)} = \sqrt{398600 \cdot \left(\frac{2}{6778} - \frac{1}{16678}\right)} = 9.681\ km/s$
  - $v_{t2} = \sqrt{\mu\left(\frac{2}{r_2} - \frac{1}{a_{trans}}\right)} = \sqrt{398600 \cdot \left(\frac{2}{26578} - \frac{1}{16678}\right)} = 2.469\ km/s$
- **Impulses & Time:**
  - $\boldsymbol{\Delta v_1} = v_{t1} - v_1 = 9.681 - 7.669 = 2.012\ km/s$
  - $\mathbf{\Delta v_2} = v_2 - v_{t2} = 3.872 - 2.469 = 1.403\ km/s$
  - $\boldsymbol{\Delta v_{total}} = \Delta v_1 + \Delta v_2 = 2.012 + 1.403 = 3.415\ km/s$

- **Transfer duration:** $t_{trans} = \pi\sqrt{\frac{a_{trans}^3}{\mu}} = \pi\sqrt{\frac{16678^3}{398600}} = 10717.591\ s$

Task 3: Bi-Elliptic vs Hohmann Comparision

Setting $R = \frac{r_2}{r_1} = 20, r_b = 40r_1$

- **Hohmann Transfer Total** Δv:
  - Transfer semi-major axis $a_h = \frac{r_1+r_2}{2} = \frac{r_1+20r_1}{2} = 10.5r_1$
  - $\Delta v_1 = v_{t1} - v_1 = \sqrt{\mu\left(\frac{2}{r_1} - \frac{1}{a_h}\right)} - \sqrt{\frac{\mu}{r_1}} = \sqrt{\mu}\left(\sqrt{\frac{2}{r_1} - \frac{1}{10.5r_1}} - \sqrt{\frac{1}{r_1}}\right) = \sqrt{\mu}\sqrt{\frac{1}{r_1}}\left(\sqrt{\frac{40}{21}} - 1\right) = 0.38v_1$
  - $\Delta v_2 = v_2 - v_{t2} = \sqrt{\frac{\mu}{r_2}} - \sqrt{\mu\left(\frac{2}{r_2} - \frac{1}{a_h}\right)} = \sqrt{\mu}\left(\sqrt{\frac{1}{20r_1}} - \sqrt{\frac{2}{20r_1} - \frac{1}{10.5r_1}}\right) = 0.15v_1$
  - $\Delta v_h = \Delta v_1 + \Delta v_2 = 0.38v_1 + 0.15v_1 = 0.53v_1$
  - Applied to LEO $(500\ km)$: $\Delta v_{Hohmann} = 0.53v_1 = 0.53\sqrt{\frac{\mu}{r_1}} = 0.53\sqrt{\frac{398600}{6378+500}} = 4.03\ km/s$
- **Bi-Elliptic Transfer Total**:
  - Ellipse 1 $(r_1 \to r_b)$:

$$a_1 = \frac{r_1 + r_b}{2} = \frac{r_1 + 40r_1}{2} = 20.5r_1$$

$$\Delta v_1 = v_{t1} - v_1 = \sqrt{\mu\left(\frac{2}{r_1} - \frac{1}{a_1}\right)} - \sqrt{\frac{\mu}{r_1}} = \sqrt{\mu}\left(\sqrt{\frac{2}{r_1} - \frac{1}{20.5r_1}} - \sqrt{\frac{1}{r_1}}\right)$$

$$= \sqrt{\mu}\sqrt{\frac{1}{r_1}}\left(\sqrt{\frac{80}{41}} - 1\right) = 0.40v_1$$

  - Ellipse 2 $(r_b \to r_2)$:

$$a_2 = \frac{r_b + r_2}{2} = \frac{40r_1 + 20r_1}{2} = 30r_1$$

$$\Delta v_2 = \sqrt{\mu\left(\frac{2}{r_b} - \frac{1}{a_2}\right)} - \sqrt{\mu\left(\frac{2}{r_b} - \frac{1}{a_1}\right)}$$

$$= \sqrt{\mu}\left(\sqrt{\frac{2}{40r_1} - \frac{1}{30r_1}} - \sqrt{\frac{2}{40r_1} - \frac{1}{20.5r_1}}\right)$$

$$= \sqrt{\mu}\sqrt{\frac{1}{r_1}}\left(\sqrt{\frac{1}{60}} - \sqrt{\frac{1}{820}}\right) = 0.09v_1$$

- Final burn (r_2):

$$\Delta v_3 = \sqrt{\frac{\mu}{r_2}} - \sqrt{\mu\left(\frac{2}{r_2} - \frac{1}{a_2}\right)} = \sqrt{\mu}\left(\sqrt{\frac{1}{20r_1}} - \sqrt{\frac{2}{20}r_1 - \frac{1}{30}r_1}\right)$$

$$= \sqrt{\mu}\sqrt{\frac{1}{r_1}}\left(\sqrt{\frac{1}{20}} - \sqrt{\frac{1}{10} - \frac{1}{30}}\right) = -0.035v_1$$

- Applied to LEO ($500\,km$): $\Delta v_{Bi-Elliptic} = |\Delta v_1| + |\Delta v_2| + |\Delta v_3| = 0.4v_1 + 0.09v_1 + 0.035v_1 = 0.525\sqrt{\frac{\mu}{r_1}} = 0.525\sqrt{\frac{398600}{6378+500}} = 4.00\ km/s$

- **Trade-Off Analysis**: Bi-elliptic saves $\approx 0.03\ km/s$ of Δv, but incurs a **massive time penalty** due to extending apogee out to $40r_1$, increasing total transfer duration from hours to days.

Task 4: Combined burn

When launching from an inclined site (e.g., $i = 51.6^o$ or 28.5^o), combining orbital circularization (Δv_2) and inclination change (Δi) into a single apogee vector burn significantly reduces overall expenditure via the Law of Cosines:

$$\Delta v_{combined} = \sqrt{v_{trans,apo}^2 + v_{GEO}^2 - 2v_{trans,apo}^2 v_{GEO}\cos(\Delta i)}$$

- **Case 1: Launch from** $i = 51.6^o$

- $v_{trans,apo} = 1.628 \frac{km}{s}, v_{GEO} = 3.075\ km/s$
- $\Delta v_{combined} = \sqrt{1.628^2 + 3.075^2 - 2 \cdot 1.628 \cdot 3.075 \cdot \cos(51.6^o)} = 2.43\ km/s$
- Comparison: Performing separate burns requires $\Delta v_2 + \Delta v_{plane} = 1.447 + 2v_{GEO} \sin\left(\frac{\Delta i}{2}\right) = 1.447 + 2 \cdot \sin\left(\frac{51.6}{2}\right) = 4.12\ km/s.$ Combined burn saves $1.69\ km/s$

- **Case 2: Launch from $i = 28.5^o$**
  - $\Delta v_{combined} = \sqrt{1.628^2 + 3.075^2 - 2 \cdot 1.628 \cdot 3.075 \cdot \cos(28.5^o)} = 1.82\ km/s$

Code for Task 2, 3, 4.

```
import numpy as np

# ========================================
# CONSTANTS & PHYSICAL PARAMETERS
# ========================================
MU = 398600   # Earth gravitational parameter (km^3/s^2)
R_E = 6378    # Earth equatorial radius (km)

def hohmann_transfer(r1, r2):
    """Calculates Hohmann transfer delta-v components and transfer duration."""
    v1 = np.sqrt(MU / r1)
    v2 = np.sqrt(MU / r2)
    a_trans = (r1 + r2) / 2.0

    v_trans_peri = np.sqrt(MU * (2.0 / r1 - 1.0 / a_trans))
    v_trans_apo = np.sqrt(MU * (2.0 / r2 - 1.0 / a_trans))

    dv1 = v_trans_peri - v1
    dv2 = v2 - v_trans_apo
    dv_total = dv1 + dv2
    t_trans_hrs = (np.pi * np.sqrt(a_trans**3 / MU)) / 3600.0

    return dv1, dv2, dv_total, t_trans_hrs, v_trans_apo, v2

# ========================================
# TASK 2: HOHMANN TRANSFERS
# ========================================
print("--- TASK 2: HOHMANN TRANSFERS ---")

# Task 2.1: LEO (500 km) to GEO
r1_500 = R_E + 500.0
r2_geo = 42164.14
dv1_geo, dv2_geo, dv_tot_geo, t_geo, v_trans_apo_geo, v_geo = hohmann_transfer(r1_500, r2_geo)

print(f"[LEO 500km -> GEO]")
print(f"  dv1: {dv1_geo:.2f} km/s | dv2: {dv2_geo:.2f} km/s | Total dv: {dv_tot_geo:.2f} km/s")
print(f"  Transfer Time: {t_geo:.2f} hours\n")

# Task 2.2: LEO (400 km) to MEO (20,200 km altitude)
r1_400 = R_E + 400.0
r2_meo = R_E + 20200.0
dv1_meo, dv2_meo, dv_tot_meo, t_meo, _, _ = hohmann_transfer(r1_400, r2_meo)

print(f"[LEO 400km -> MEO 20,200km]")
print(f"  dv1: {dv1_meo:.2f} km/s | dv2: {dv2_meo:.2f} km/s | Total dv: {dv_tot_meo:.2f} km/s")
print(f"  Transfer Time: {t_meo:.2f} hours\n")
```

```
# ==========================================
# TASK 3: BI-ELLIPTIC COMPARISON (R = 20, r_b = 40*r1)
# ==========================================
print("--- TASK 3: BI-ELLIPTIC VS HOHMANN (R = 20) ---")

r1_norm = 1.0
r2_norm = 20.0
rb_norm = 40.0
v1_phys = np.sqrt(MU / r1_500)

# Hohmann for R = 20
a_h = (r1_norm + r2_norm) / 2.0
v_tp1_h = np.sqrt(2.0/r1_norm - 1.0/a_h)
v_tp2_h = np.sqrt(2.0/r2_norm - 1.0/a_h)
v2_h = np.sqrt(1.0/r2_norm)

dv_h_tot = (v_tp1_h - 1.0) + (v2_h - v_tp2_h)
dv_h_phys = dv_h_tot * v1_phys

# Bi-elliptic for R = 20, r_b = 40
a1_b = (r1_norm + rb_norm) / 2.0
v_tp1_b = np.sqrt(2.0/r1_norm - 1.0/a1_b)
v_tpb1 = np.sqrt(2.0/rb_norm - 1.0/a1_b)

a2_b = (r2_norm + rb_norm) / 2.0
v_tpb2 = np.sqrt(2.0/rb_norm - 1.0/a2_b)
v_tp2_b = np.sqrt(2.0/r2_norm - 1.0/a2_b)

dv_bi_tot = abs(v_tp1_b - 1.0) + abs(v_tpb2 - v_tpb1) + abs(v2_h - v_tp2_b)
dv_bi_phys = dv_bi_tot * v1_phys

print(f"Hohmann Total (R=20):     {dv_h_phys:.2f} km/s")
print(f"Bi-elliptic Total (r_b=40): {dv_bi_phys:.2f} km/s")
print(f"Delta-v Savings:          {dv_h_phys - dv_bi_phys:.2f} km/s (Bi-elliptic wins)\n")

# ==========================================
# TASK 4: COMBINED BURN (LEO 500km -> GEO with i = 51.6 deg)
# ==========================================
print("--- TASK 4: COMBINED APOGEE BURN ---")

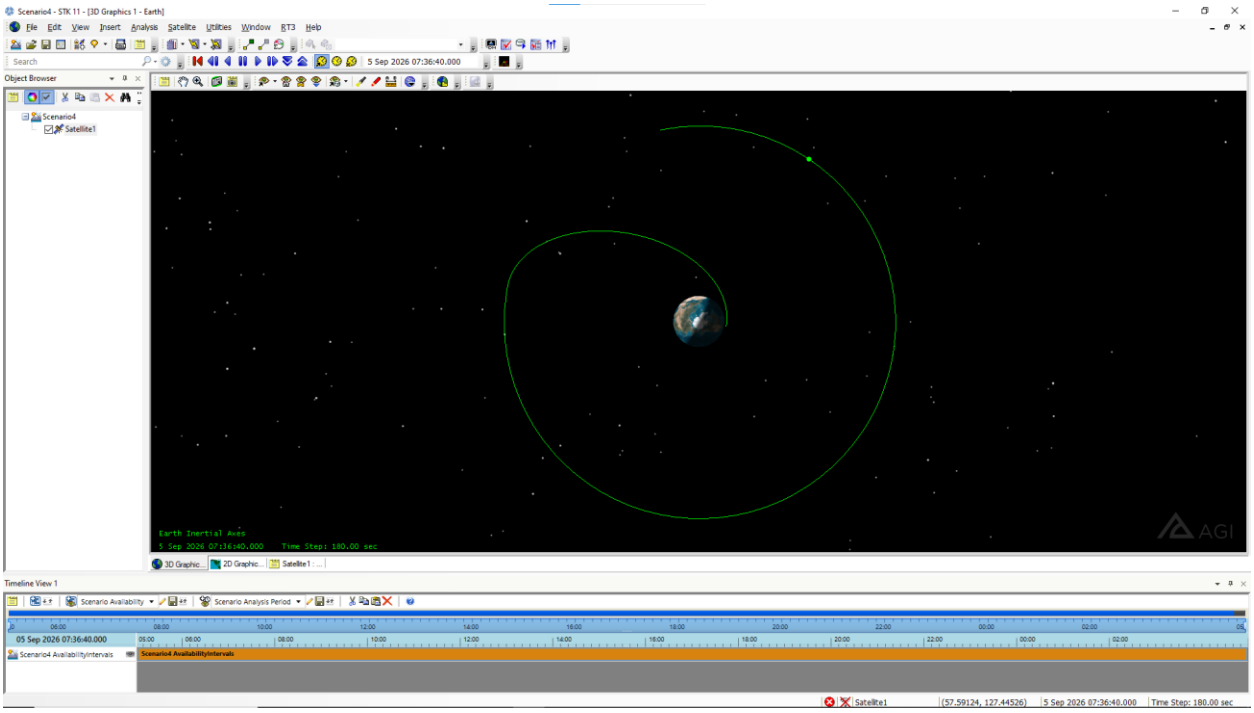
inc_deg = 51.6
inc_rad = np.radians(inc_deg)

# Combined Vector Burn via Law of Cosines
dv_combined = np.sqrt(v_trans_apo_geo**2 + v_geo**2 - 2.0 * v_trans_apo_geo * v_geo *
np.cos(inc_rad))

# Separate In-Plane Burn + Plane Change
dv_plane_only = 2.0 * v_geo * np.sin(inc_rad / 2.0)
dv_separate = dv2_geo + dv_plane_only

print(f"Combined Apogee Burn Cost: {dv_combined:.2f} km/s")
print(f"Separate Burns Cost:       {dv_separate:.2f} km/s")
print(f"Savings via Vector Combined Burn: {dv_separate - dv_combined:.2f} km/s")
```

Task 5: STK



Target Sequence.Differential Corrector: Finished: *CONVERGED* in 0 iterations. *Constraints Met*

Control	New Value	Last Update	Constraint	Desired	Achieved	Difference	Tolerance
...ulsiveMnvr.Cartesian.XSDU	2465.81 m/s	0 m/sec	...agate1 : Altitude Of Apoapsis	42164 km	42164.1 k	0.059072 k	0.1 km
...ulsiveMnvr.Cartesian.XSDU	1437.51 m/s	0 m/sec	Maneuver2 : Eccentricity	0	2.3203e-0	2.3203e-06	0.0001